Dr. Mahalingam College of Engineering and Technology, Pollachi-642003

Department of Information Technology

Academic Year: 2025 - 2026

TRL and SDG Certificate

Project Title : Student Attendance management system

Course Code & Name : 23ITI402 Database Management Systems

Department and Semester: Information Technology & IV SEM

Technology Readiness Level (TRL) of the Project : _____

Sustainability Development Goals (SDG)-Goal Name : _____

S.No.	Names and Roll Numbers of Student	Signature of the Student
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Preparation (25 marks)	
Interpretation (30 marks)	
Output & Result (15 marks)	
Viva (5 marks)	
Total (75 marks)	
Signature of the faculty incharge	

Objective of the Project

The primary objective of this project is to design and develop an automated Student Attendance and Performance Management System using Java Swing and MySQL, specifically aimed at overcoming the limitations of traditional manual record-keeping methods. By digitizing the attendance process, the system replaces paper-based registers with a streamlined interface that allows for quick marking of student status while automatically capturing the current date and time to ensure data integrity and prevent back-dated entries. Beyond attendance tracking, the project integrates a robust academic performance module that stores subject-wise marks and automates total score calculations, thereby significantly reducing the possibility of human error. The system is built on a relational database architecture, providing the administrator with full control through complete CRUD (Create, Read, Update, Delete) operations, while the user-friendly graphical interface features a dynamic toggle mechanism to efficiently switch between attendance and performance views. Ultimately, this project aims to provide a reliable, scalable, and secure digital platform that enhances the efficiency of educational data management.

Entire Project Code

```
import javax.swing.*;

import javax.swing.border.EmptyBorder;

import java.awt.*;

import java.awt.event.*;


public class Login_UI extends JFrame {

    private JTextField tUser;

    private JPasswordField tPass;

    private JButton bLogin, bClear, bExit;

    private JCheckBox showPassword;

    private JProgressBar progressBar;
```

```
private JLabel statusLabel;

public Login_UI() {
    setTitle("Enterprise Secure Access - v3.0");
    setSize(1100, 800);
    setDefaultCloseOperation(EXIT_ON_CLOSE);
    setLocationRelativeTo(null);
    setResizable(false);

    JPanel container = new JPanel(new BorderLayout());
    container.setBackground(new Color(235, 240, 245));
    add(container);

    JPanel header = new JPanel();
    header.setBackground(new Color(15, 45, 80));
    header.setPreferredSize(new Dimension(1100, 120));
    header.setLayout(new GridBagLayout());

    JLabel mainTitle = new JLabel("SECURE INFRASTRUCTURE LOGIN");
    mainTitle.setFont(new Font("Verdana", Font.BOLD, 36));
    mainTitle.setForeground(Color.WHITE);
    header.add(mainTitle);
    container.add(header, BorderLayout.NORTH);

    JPanel centerPanel = new JPanel(new GridBagLayout());
    centerPanel.setOpaque(false);
    GridBagConstraints gbc = new GridBagConstraints();
```

```
JPanel card = new JPanel(new GridBagLayout());
card.setBackground(Color.WHITE);
card.setBorder(BorderFactory.createCompoundBorder(
    BorderFactory.createLineBorder(new Color(180, 180, 180), 1),
    BorderFactory.createEmptyBorder(30, 40, 30, 40)
));

GridBagConstraints cgbc = new GridBagConstraints();
cgbc.insets = new Insets(12, 12, 12, 12);
cgbc.fill = GridBagConstraints.HORIZONTAL;

JLabel lIcon = new JLabel("AUTHENTICATION REQUIRED", JLabel.CENTER);
lIcon.setFont(new Font("Impact", Font.PLAIN, 28));
lIcon.setForeground(new Color(50, 50, 50));
cgbc.gridx = 0; cgbc.gridy = 0; cgbc.gridwidth = 2;
card.add(lIcon, cgbc);

cgbc.gridwidth = 1; cgbc.gridy = 1;
JLabel lUser = new JLabel("OPERATOR ID");
lUser.setFont(new Font("SansSerif", Font.BOLD, 13));
card.add(lUser, cgbc);

cgbc.gridy = 2;
tUser = new JTextField();
tUser.setFont(new Font("Segoe UI", Font.PLAIN, 20));
tUser.setPreferredSize(new Dimension(380, 45));
```

```
tUser.setBorder(BorderFactory.createLineBorder(new Color(200, 200, 200)));
card.add(tUser, cgbc);

cgbc.gridy = 3;
JLabel lPass = new JLabel("ACCESS KEY");
lPass.setFont(new Font("SansSerif", Font.BOLD, 13));
card.add(lPass, cgbc);

cgbc.gridy = 4;
tPass = new JPasswordField();
tPass.setFont(new Font("Segoe UI", Font.PLAIN, 20));
tPass.setPreferredSize(new Dimension(380, 45));
tPass.setBorder(BorderFactory.createLineBorder(new Color(200, 200, 200)));
card.add(tPass, cgbc);

cgbc.gridy = 5;
showPassword = new JCheckBox("Show Plaintext Password");
showPassword.setBackground(Color.WHITE);
showPassword.setFocusPainted(false);
card.add(showPassword, cgbc);

cgbc.gridy = 6;
progressBar = new JProgressBar();
progressBar.setIndeterminate(true);
progressBar.setVisible(false);
progressBar.setForeground(new Color(0, 120, 215));
card.add(progressBar, cgbc);
```

```
cgbc.gridy = 7;
JPanel actionPanel = new JPanel(new GridLayout(1, 2, 20, 0));
actionPanel.setOpaque(false);

bLogin = createStyledButton("AUTHORIZE", new Color(40, 160, 80));
bClear = createStyledButton("RESET FIELDS", new Color(210, 80, 40));

actionPanel.add(bLogin);
actionPanel.add(bClear);
card.add(actionPanel, cgbc);

gbc.gridx = 0; gbc.gridy = 0;
centerPanel.add(card, gbc);
container.add(centerPanel, BorderLayout.CENTER);

JPanel footer = new JPanel(new BorderLayout());
footer.setBackground(new Color(220, 220, 220));
footer.setPreferredSize(new Dimension(1100, 50));
footer.setBorder(new EmptyBorder(0, 20, 0, 20));

statusLabel = new JLabel("System Status: Ready | Connection: Encrypted (AES-256)");
statusLabel.setFont(new Font("Monospaced", Font.BOLD, 12));
footer.add(statusLabel, BorderLayout.WEST);

bExit = new JButton("SHUTDOWN SYSTEM");
```

```
        bExit.setFont(new Font("Monospaced", Font.BOLD, 12));
        bExit.setForeground(new Color(150, 0, 0));
        bExit.setBorderPainted(false);
        bExit.setContentAreaFilled(false);
        bExit.setCursor(new Cursor(Cursor.HAND_CURSOR));
        footer.add(bExit, BorderLayout.EAST);

        container.add(footer, BorderLayout.SOUTH);

        showPassword.addActionListener(e -> {
            tPass.setEchoChar(showPassword.isSelected() ? (char) 0 : '•');
        });

        bLogin.addActionListener(e -> {
            String user = tUser.getText();
            String pass = new String(tPass.getPassword());

            if (user.isEmpty() || pass.isEmpty()) {
                JOptionPane.showMessageDialog(this, "Empty fields detected!", "Data Error",
JOptionPane.WARNING_MESSAGE);
                return;
            }

            progressBar.setVisible(true);
            statusLabel.setText("System Status: Verifying Credentials...");

            Timer timer = new Timer(2000, evt -> {
```

```
            progressBar.setVisible(false);
            if (user.equals("admin") && pass.equals("password")) {
                JOptionPane.showMessageDialog(this, "Access Granted. Initializing Dashboard.", "Success", JOptionPane.INFORMATION_MESSAGE);
                dispose();
            } else {
                statusLabel.setText("System Status: Access Denied");
                JOptionPane.showMessageDialog(this, "Critical: Invalid Credentials", "Security Error", JOptionPane.ERROR_MESSAGE);
            }
        });
        timer.setRepeats(false);
        timer.start();
    });

    bClear.addActionListener(e -> {
        tUser.setText("");
        tPass.setText("");
        progressBar.setVisible(false);
        statusLabel.setText("System Status: Ready");
        tUser.requestFocus();
    });

    bExit.addActionListener(e -> System.exit(0));

    setVisible(true);
  }
```

```
    private JButton createStyledButton(String text, Color bg) {
        JButton btn = new JButton(text);
        btn.setFont(new Font("Segoe UI", Font.BOLD, 16));
        btn.setBackground(bg);
        btn.setForeground(Color.WHITE);
        btn.setFocusPainted(false);
        btn.setCursor(new Cursor(Cursor.HAND_CURSOR));
        btn.setPreferredSize(new Dimension(150, 50));

        btn.addMouseListener(new MouseAdapter() {
            public void mouseEntered(MouseEvent e) { btn.setBackground(bg.brighter()); }
            public void mouseExited(MouseEvent e) { btn.setBackground(bg); }
        });
        return btn;
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> {
            try {
UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName());
            } catch (Exception ignored) {}
            new Login_UI();
        });
    }
}
```

DB_connection.java

```
import java.sql.*;
import java.util.Properties;

public class DBConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/attandance";
    private static final String USER = "root";
    private static final String PASS = "dbms";
    private static Connection connection = null;

    public static Connection getConnection() throws SQLException {
        if (connection == null || connection.isClosed()) {
            try {
                // Loading the MySQL JDBC Driver explicitly for better compatibility
                Class.forName("com.mysql.cj.jdbc.Driver");

                Properties props = new Properties();
                props.setProperty("user", USER);
                props.setProperty("password", PASS);
                props.setProperty("useSSL", "false");
                props.setProperty("serverTimezone", "UTC");
                props.setProperty("allowPublicKeyRetrieval", "true");

                connection = DriverManager.getConnection(URL, props);
                System.out.println("LOG: Database Connection Established Successfully.");
            } catch (ClassNotFoundException e) {
```

```
            System.err.println("CRITICAL: MySQL Driver Not Found!");
        }
    }
    return connection;
}

public static void checkServerStatus() {
    try (Connection conn = getConnection()) {
        if (conn.isValid(5)) {
            DatabaseMetaData meta = conn.getMetaData();
            System.out.println("SERVER INFO: " + meta.getDatabaseProductVersion());
            System.out.println("DRIVER INFO: " + meta.getDriverName());
        }
    } catch (SQLException e) {
        System.err.println("ALERT: Database Server is Offline or Unreachable!");
    }
}

public static void closeConnection() {
    try {
        if (connection != null && !connection.isClosed()) {
            connection.close();
            System.out.println("LOG: Database Connection Closed Safely.");
        }
    } catch (SQLException e) {
        e.printStackTrace();
    }
```

```
    }
}
```

ATTENDANCE_UI.java

```
import javax.swing.*;
import javax.swing.table.*;
import javax.swing.border.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;
import java.time.LocalDate;
import java.text.MessageFormat;

public class Student_UI extends JFrame {
    private JTextField tRoll, tName, tM1, tM2, tSearch, tEmail;
    private JComboBox<String> comboStatus, comboDept;
    private JTable attTable, markTable;
    private DefaultTableModel attModel, markModel;
    private JPanel markPanel, attPanel, statsPanel;
    private JButton bToggle, bSave, bDel, bClear, bMarkSave, bSearch, bPrint, bUpdate;
    private JLabel statusInfo, lblTotalStudents, lblPresentToday;

    public Student_UI() {
        setTitle("ELITE ACADEMIC ERP SYSTEM - VERSION 5.0");
        setSize(1300, 900);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLayout(new BorderLayout(15, 15));
```

```
        initMenuBar();

        // --- SIDE NAVIGATION BAR ---
        JPanel sideNav = new JPanel(new GridLayout(10, 1, 10, 10));
        sideNav.setBackground(new Color(33, 47, 61));
        sideNav.setBorder(BorderFactory.createEmptyBorder(25, 15, 25, 15));

        bToggle = createNavButton("CHANGE MODULE");
        bClear = createNavButton("RESET ALL");
        bPrint = createNavButton("GENERATE REPORT");
        JButton bLogout = createNavButton("SYSTEM LOGOUT");

        sideNav.add(new JLabel("CONTROL CENTER",
JLabel.CENTER)).setForeground(Color.CYAN);
        sideNav.add(bToggle);
        sideNav.add(bClear);
        sideNav.add(bPrint);
        sideNav.add(new JSeparator());
        sideNav.add(bLogout);
        add(sideNav, BorderLayout.WEST);

        // --- MAIN CONTENT AREA ---
        JPanel mainContent = new JPanel();
        mainContent.setLayout(new BoxLayout(mainContent, BoxLayout.Y_AXIS));
        mainContent.setBackground(new Color(245, 245, 250));
        add(new JScrollPane(mainContent), BorderLayout.CENTER);
```

```
// --- STATISTICS PANEL (New Feature) ---
statsPanel = new JPanel(new GridLayout(1, 3, 20, 0));
statsPanel.setMaximumSize(new Dimension(1600, 100));
statsPanel.setOpaque(false);
statsPanel.add(createStatCard("Total Records", lblTotalStudents = new
JLabel("0")));
statsPanel.add(createStatCard("Present Today", lblPresentToday = new
JLabel("0")));
statsPanel.add(createStatCard("Server Status", new JLabel("Online")));
mainContent.add(statsPanel);
mainContent.add(Box.createRigidArea(new Dimension(0, 20)));

// --- ATTENDANCE SECTION ---
attPanel = new JPanel(new BorderLayout(10, 10));
attPanel.setBackground(Color.WHITE);
attPanel.setBorder(new TitledBorder(new LineBorder(Color.GRAY, 2),
"ATTENDANCE REGISTRY", 0, 0, new Font("Dialog", Font.BOLD, 20)));

JPanel attHeader = new JPanel(new GridBagLayout());
attHeader.setOpaque(false);
GridBagConstraints gbc = new GridBagConstraints();
gbc.insets = new Insets(8, 8, 8, 8);
gbc.fill = GridBagConstraints.HORIZONTAL;

tRoll = new JTextField(8);
tName = new JTextField(12);
tEmail = new JTextField(12);
```

```
        comboDept = new JComboBox<>(new String[]{"Computer Science", "IT",
"Electronics", "Mechanical"});

        comboStatus = new JComboBox<>(new String[]{"Present", "Absent", "Late",
"Medical Leave"});


        gbc.gridx = 0; gbc.gridy = 0; attHeader.add(new JLabel("Roll No:"), gbc);

        gbc.gridx = 1; attHeader.add(tRoll, gbc);

        gbc.gridx = 2; attHeader.add(new JLabel("Student Name:"), gbc);

        gbc.gridx = 3; attHeader.add(tName, gbc);

        gbc.gridx = 4; attHeader.add(new JLabel("Department:"), gbc);

        gbc.gridx = 5; attHeader.add(comboDept, gbc);


        JPanel attButtons = new JPanel(new FlowLayout(FlowLayout.RIGHT));

        bSave = createActionButton("SAVE RECORD", new Color(40, 167, 69));

        bUpdate = createActionButton("UPDATE", new Color(255, 193, 7));

        bDel = createActionButton("DELETE", new Color(220, 53, 69));

        tSearch = new JTextField(12);

        bSearch = new JButton("Search Roll");


        attButtons.add(new JLabel("Filter: ")); attButtons.add(tSearch);

        attButtons.add(bSearch); attButtons.add(bSave); attButtons.add(bUpdate);
attButtons.add(bDel);


        gbc.gridx = 0; gbc.gridy = 1; gbc.gridwidth = 6;

        attHeader.add(attButtons, gbc);

        attPanel.add(attHeader, BorderLayout.NORTH);
```

```
        attModel = new DefaultTableModel(new String[]{"Roll No", "Name", "Dept",
"Date", "Status"}, 0);

        attTable = new JTable(attModel);

        customizeTable(attTable);

        attPanel.add(new JScrollPane(attTable), BorderLayout.CENTER);

        mainContent.add(attPanel);


        // --- MARK SECTION ---

        markPanel = new JPanel(new BorderLayout(10, 10));

        markPanel.setBackground(Color.WHITE);

        markPanel.setVisible(false);

        markPanel.setBorder(new TitledBorder(new LineBorder(Color.BLUE, 2),
"ACADEMIC MARKSHEET", 0, 0, new Font("Dialog", Font.BOLD, 20)));


        JPanel markInput = new JPanel(new FlowLayout(FlowLayout.LEFT, 20, 15));

        tM1 = new JTextField(5); tM2 = new JTextField(5);

        bMarkSave = createActionButton("SUBMIT MARKS", new Color(0, 123, 255));


        markInput.add(new JLabel("Internal Marks:")); markInput.add(tM1);

        markInput.add(new JLabel("External Marks:")); markInput.add(tM2);

        markInput.add(bMarkSave);

        markPanel.add(markInput, BorderLayout.NORTH);


        markModel = new DefaultTableModel(new String[]{"Roll", "Name", "Internal",
"External", "Total", "Grade", "Status"}, 0);

        markTable = new JTable(markModel);

        customizeTable(markTable);

        markPanel.add(new JScrollPane(markTable), BorderLayout.CENTER);
```

```
        mainContent.add(markPanel);

        // --- FOOTER ---
        JPanel footer = new JPanel(new BorderLayout());
        footer.setBackground(new Color(230, 230, 230));
        statusInfo = new JLabel(" System Ready | User: Admin | Logged in at: " +
LocalDate.now());
        footer.add(statusInfo, BorderLayout.WEST);
        add(footer, BorderLayout.SOUTH);

        // --- LISTENERS ---
        bToggle.addActionListener(e -> {
            boolean isVis = markPanel.isVisible();
            markPanel.setVisible(!isVis);
            bToggle.setText(isVis ? "VIEW MARKS" : "VIEW ATTENDANCE");
            revalidate(); repaint();
        });

        bPrint.addActionListener(e -> printReport());
        bSave.addActionListener(e -> validateAndSave());
        bMarkSave.addActionListener(e -> saveMarks());
        bDel.addActionListener(e -> deleteRecord());
        bClear.addActionListener(e -> clearInputs());
        bLogout.addActionListener(e -> System.exit(0));

        refresh();
        setLocationRelativeTo(null);
```

```
        setVisible(true);
    }

    private void initMenuBar() {
        JMenuBar mb = new JMenuBar();
        JMenu file = new JMenu("File");
        JMenuItem export = new JMenuItem("Export to PDF");
        JMenuItem exit = new JMenuItem("Exit");
        file.add(export); file.add(exit);
        mb.add(file);
        setJMenuBar(mb);
    }

    private JPanel createStatCard(String title, JLabel value) {
        JPanel p = new JPanel(new BorderLayout());
        p.setBackground(Color.WHITE);
        p.setBorder(BorderFactory.createCompoundBorder(new LineBorder(new
Color(200, 200, 200), 1), new EmptyBorder(10, 10, 10, 10)));
        JLabel t = new JLabel(title, JLabel.CENTER);
        t.setFont(new Font("Arial", Font.PLAIN, 14));
        value.setFont(new Font("Arial", Font.BOLD, 22));
        value.setHorizontalAlignment(JLabel.CENTER);
        p.add(t, BorderLayout.NORTH);
        p.add(value, BorderLayout.CENTER);
        return p;
    }
```

```
private JButton createNavButton(String text) {
    JButton b = new JButton(text);
    b.setForeground(Color.WHITE);
    b.setBackground(new Color(52, 73, 94));
    b.setFocusPainted(false);
    b.setBorder(new LineBorder(Color.DARK_GRAY));
    return b;
}

private JButton createActionButton(String text, Color bg) {
    JButton b = new JButton(text);
    b.setBackground(bg);
    b.setForeground(Color.WHITE);
    b.setFont(new Font("Arial", Font.BOLD, 12));
    return b;
}

private void customizeTable(JTable table) {
    table.setRowHeight(30);
    table.getTableHeader().setFont(new Font("Arial", Font.BOLD, 14));
    table.setSelectionBackground(new Color(174, 214, 241));
}

private void validateAndSave() {
    if(tRoll.getText().isEmpty() || tName.getText().isEmpty()) {
        JOptionPane.showMessageDialog(this, "Roll No and Name are mandatory!");
        return;
```

```
        }
        saveAttendance();
    }

    private void printReport() {
        try {
            MessageFormat header = new MessageFormat("Student Report - " +
LocalDate.now());
            MessageFormat footer = new MessageFormat("Page {0,number,integer}");
            attTable.print(JTable.PrintMode.FIT_WIDTH, header, footer);
        } catch (Exception e) { JOptionPane.showMessageDialog(this, "Print Failed: " +
e.getMessage()); }
    }

    private void saveAttendance() {
        try (Connection c = DBConnection.getConnection();
             PreparedStatement p = c.prepareStatement("INSERT INTO daily_attendance
(roll_no, name, dept, date, status) VALUES (?,?,?,?,?)")) {
            p.setInt(1, Integer.parseInt(tRoll.getText()));
            p.setString(2, tName.getText());
            p.setString(3, comboDept.getSelectedItem().toString());
            p.setDate(4, Date.valueOf(LocalDate.now()));
            p.setString(5, comboStatus.getSelectedItem().toString());
            p.executeUpdate();
            refresh();
            statusInfo.setText(" Last Action: Record Saved successfully!");
        } catch (Exception ex) { ex.printStackTrace(); }
    }
```

```
    private void saveMarks() {
        try (Connection c = DBConnection.getConnection();
            PreparedStatement p = c.prepareStatement("INSERT INTO subject_marks
VALUES (?,?,?,?,?)")) {
            int m1 = Integer.parseInt(tM1.getText());
            int m2 = Integer.parseInt(tM2.getText());
            int total = m1 + m2;
            p.setInt(1, Integer.parseInt(tRoll.getText()));
            p.setString(2, tName.getText());
            p.setInt(3, m1);
            p.setInt(4, m2);
            p.setInt(5, total);
            p.executeUpdate();
            refresh();
        } catch (Exception ex) { ex.printStackTrace(); }
    }

    private void deleteRecord() {
        int row = attTable.getSelectedRow();
        if (row == -1) return;
        int confirm = JOptionPane.showConfirmDialog(this, "Are you sure you want to
delete this?");
        if (confirm == JOptionPane.YES_OPTION) {
            String roll = attModel.getValueAt(row, 0).toString();
            try (Connection c = DBConnection.getConnection();
                PreparedStatement p = c.prepareStatement("DELETE FROM daily_attendance
WHERE roll_no = ?")) {
```

```
            p.setString(1, roll);
            p.executeUpdate();
            refresh();
        } catch (Exception ex) { ex.printStackTrace(); }
    }
}

private void clearInputs() {
    tRoll.setText(""); tName.setText(""); tM1.setText(""); tM2.setText(""); tEmail.setText("");
    statusInfo.setText(" System Status: Inputs Cleared");
}

private void refresh() {
    attModel.setRowCount(0);
    markModel.setRowCount(0);
    int total = 0, present = 0;
    try (Connection c = DBConnection.getConnection()) {
        ResultSet r1 = c.createStatement().executeQuery("SELECT * FROM daily_attendance");
        while(r1.next()) {
            attModel.addRow(new Object[]{r1.getInt(1), r1.getString(2), r1.getString(3), r1.getDate(4), r1.getString(5)});
            total++;
            if(r1.getString(5).equals("Present")) present++;
        }
        lblTotalStudents.setText(String.valueOf(total));
        lblPresentToday.setText(String.valueOf(present));
```

```
        ResultSet r2 = c.createStatement().executeQuery("SELECT * FROM
subject_marks");

        while(r2.next()) {

            int sum = r2.getInt(5);

            String grd = (sum > 90) ? "O" : (sum > 75) ? "A+" : "B";

            String res = (sum >= 40) ? "PASS" : "FAIL";

            markModel.addRow(new Object[]{r2.getInt(1), r2.getString(2), r2.getInt(3),
r2.getInt(4), sum, grd, res});

        }

      } catch (Exception e) { e.printStackTrace(); }

   }

}
```

Database view

The database consists of two tables:

1 The structure of table

```
mysql> desc daily_attendance;
+-----------------+--------------+------+-----+---------+-------+
| Field           | Type         | Null | Key | Default | Extra |
+-----------------+--------------+------+-----+---------+-------+
| roll_no         | int          | NO   | PRI | NULL    |       |
| student_name    | varchar(100) | YES  |     | NULL    |       |
| attendance_date | date         | YES  |     | NULL    |       |
| status          | varchar(10)  | YES  |     | NULL    |       |
+-----------------+--------------+------+-----+---------+-------+
4 rows in set (0.03 sec)
```

```
mysql> desc subject_marks;
+--------------+--------------+------+-----+---------+-------+
| Field        | Type         | Null | Key | Default | Extra |
+--------------+--------------+------+-----+---------+-------+
| roll_no      | int          | NO   | PRI | NULL    |       |
| student_name | varchar(100) | YES  |     | NULL    |       |
| subject_1    | int          | YES  |     | NULL    |       |
| subject_2    | int          | YES  |     | NULL    |       |
| total        | int          | YES  |     | NULL    |       |
+--------------+--------------+------+-----+---------+-------+
```

```
mysql> select * from daily_attendance;
+---------+--------------+-----------------+--------+
| roll_no | student_name | attendance_date | status |
+---------+--------------+-----------------+--------+
|      23 | kumar        | 2026-04-17      | Yes    |
|      29 | ram          | 2026-04-17      | Yes    |
|      38 | thamarai     | 2026-04-17      | Yes    |
|      61 | vimal        | 2026-04-17      | Yes    |
|      99 | vishnu       | 2026-04-17      | No     |
+---------+--------------+-----------------+--------+
5 rows in set (0.00 sec)
```

```
mysql> select * from subject_marks;
+---------+--------------+-----------+-----------+-------+
| roll_no | student_name | subject_1 | subject_2 | total |
+---------+--------------+-----------+-----------+-------+
|      23 | kumar        |        85 |        90 |   175 |
|      29 | ram          |        70 |        75 |   145 |
|      38 | thamarai     |        99 |        99 |   198 |
|      61 | vimal        |        90 |        92 |   182 |
|      99 | vishnu       |        88 |        63 |   151 |
+---------+--------------+-----------+-----------+-------+
5 rows in set (0.00 sec)
```

Output screenshots

Figure 1: Login Screen

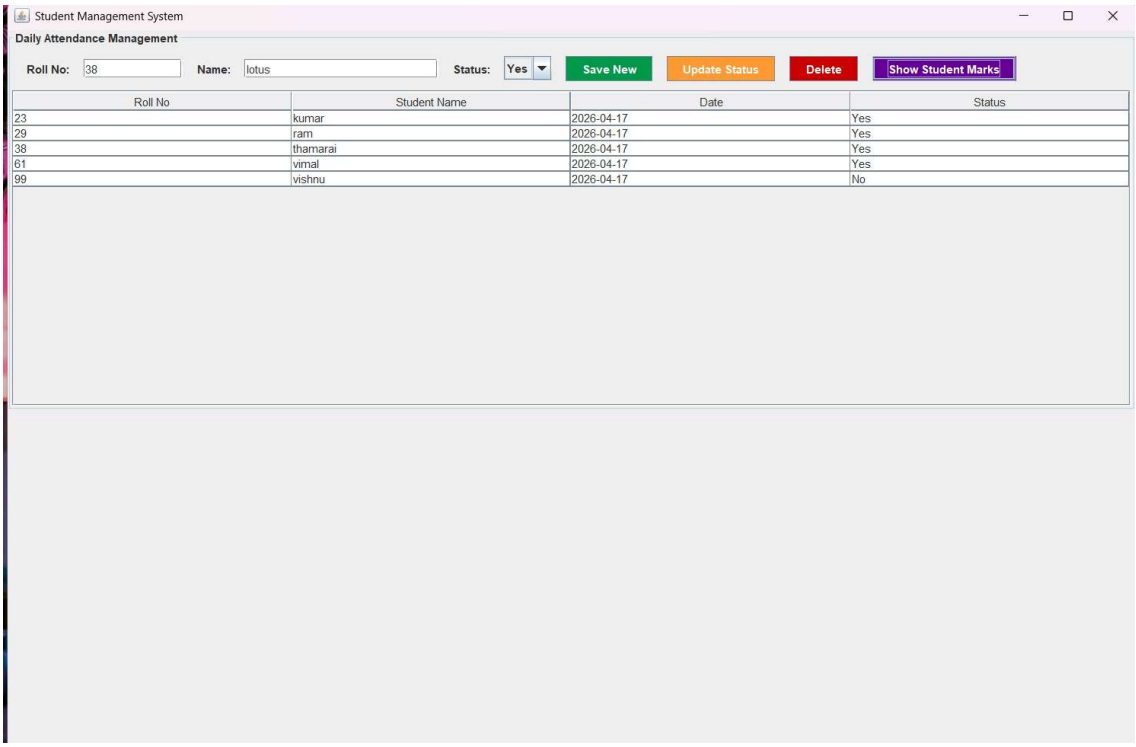


Figure 2: Dashboard

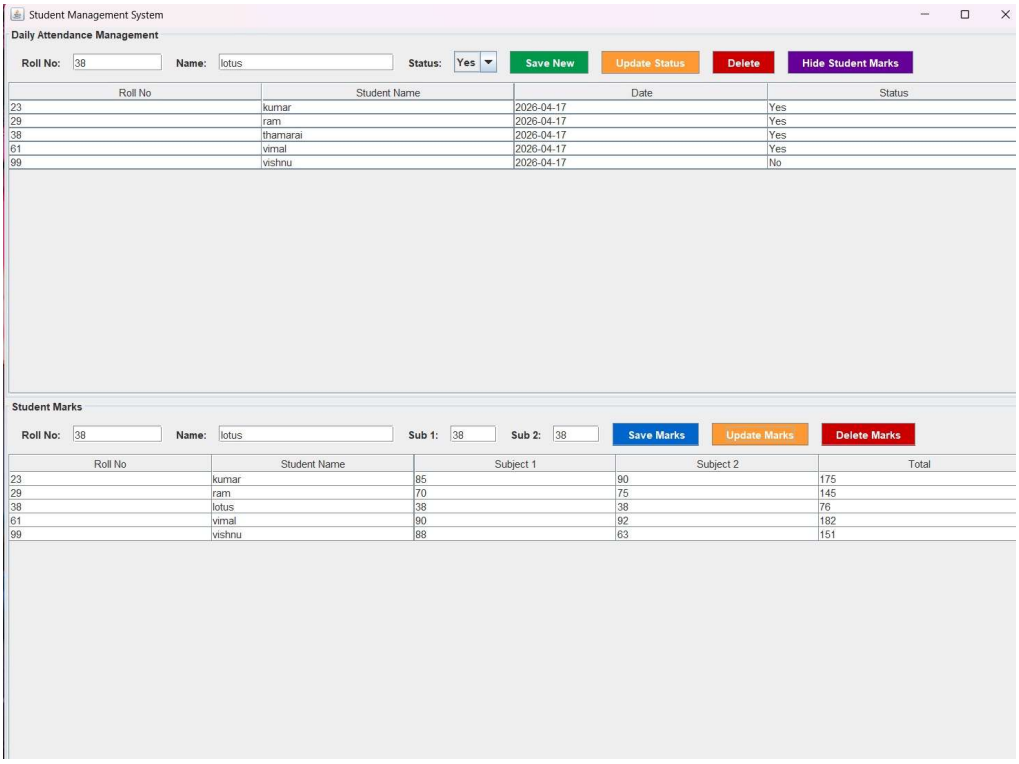


Figure 3: Hide student mark

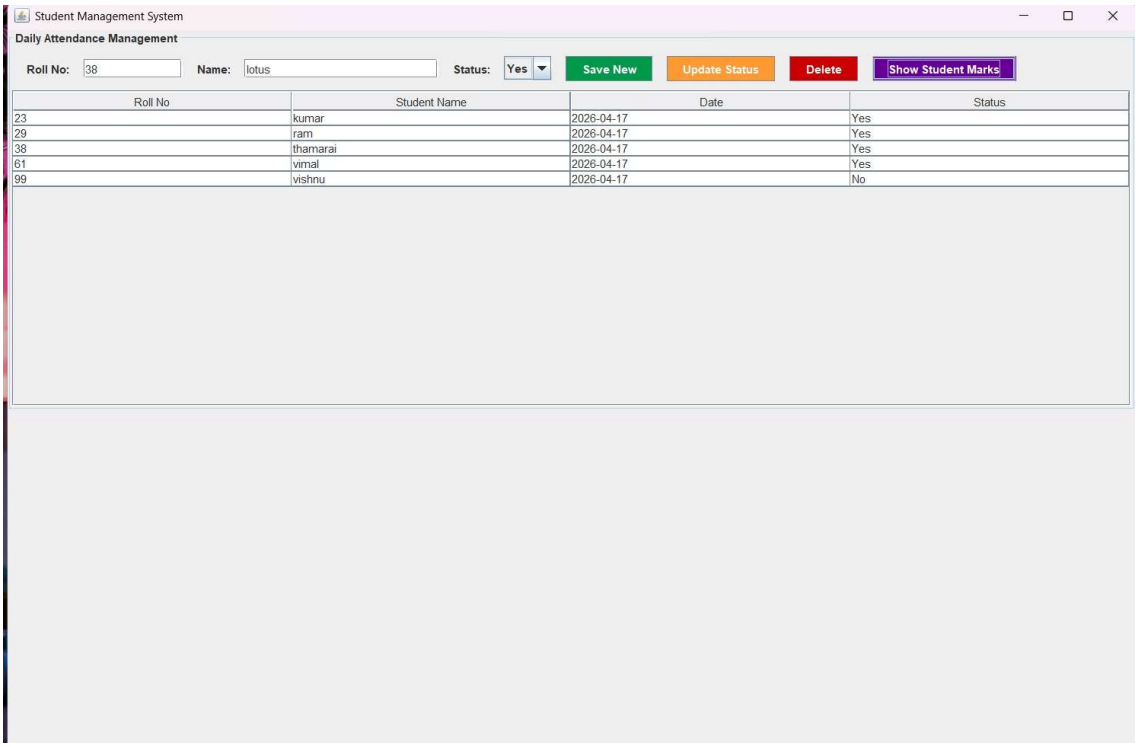
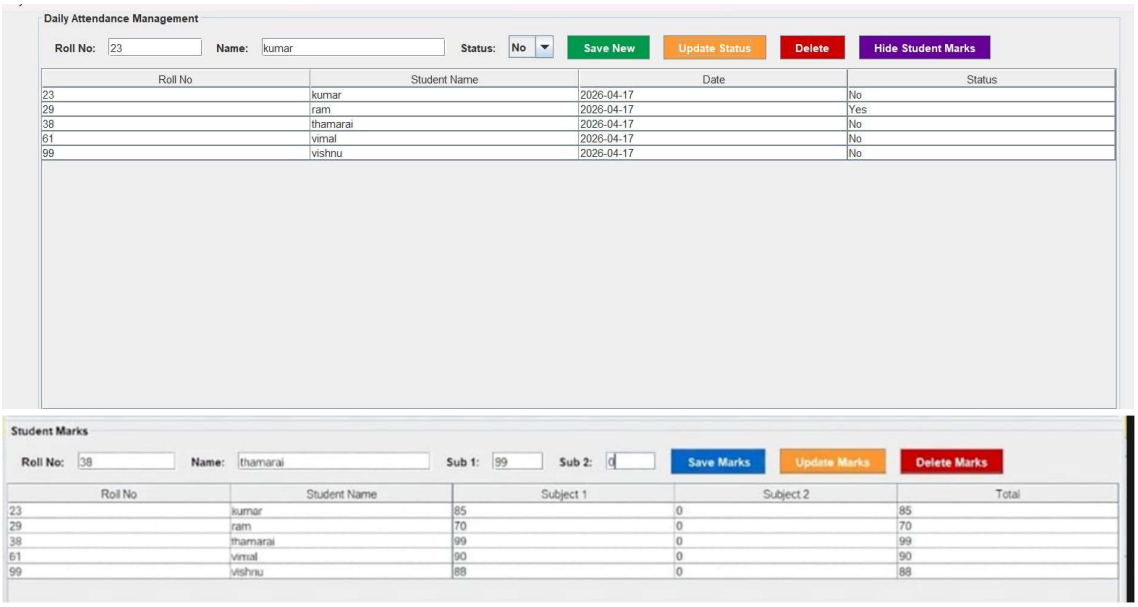


Figure 4: After Update



Result

The Student Attendance system was successfully implemented using Java Swing for the user interface and MySQL for database management.

The system provides functionalities such as student attendance marking, daily status updates, and record deletion. It also includes features to store academic marks, calculate total scores automatically, and a dynamic toggle interface to manage attendance and performance data efficiently.

All modules were tested and verified, and the system produced accurate and reliable results. Hence, the project fulfills its intended objectives successfully.